



Enterobius (Pinworm) — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. What is the common name of *Enterobius vermicularis*?

EASY

- A. Hookworm
- B. Beef tapeworm
- C. Pork tapeworm
- D. Pinworm
- E. Whipworm

2. Which age group is most commonly infected with pinworm?

EASY

- A. Under 1 year
- B. 1–2 years
- C. 20–40 years
- D. Over 60 years
- E. 5–14 years

3. Which specimen is used to demonstrate pinworm eggs?

EASY

- A. Perianal skin (cellophane tape)
- B. Sputum
- C. Urine
- D. Blood film
- E. Bile aspirate

4. Which symptom is most characteristic of pinworm infection?

EASY

- A. Perianal itching, worse at night
- B. Ground itch on the feet
- C. Hemoptysis
- D. Painless jaundice
- E. Melena

5. A mother asks why her son's itching is worst at night. What triggers the gravid female to migrate to the perianal region?

MEDIUM

- A. Bright light
- B. Loud noise
- C. Low temperature and an aerobic environment
- D. High blood glucose
- E. Gastric acid

6. Eggs freshly laid on a child's skin become infectious after about:

MEDIUM

- A. 1 hour
- B. 24 hours
- C. 6 hours
- D. 5 days
- E. 2 weeks

7. Which combination best describes the egg on microscopy?

MEDIUM

- A. Round, bile-stained, sinks in salt
- B. Biconvex, thin shell, 7–8 blastomeres
- C. Planoconvex, thick transparent shell, not bile-stained, floats in saturated salt
- D. Operculated with a spine
- E. Colourless, thin shell, 60 × 40 µm with clear space

8. A worm from a child's perianal skin shows a double-bulb esophagus. Which statement is correct?

MEDIUM

- A. It is shared with hookworms
- B. It is typical of tapeworms
- C. It marks the male only
- D. It appears in eggs
- E. This feature is unique to *E. vermicularis*

9. Which dosing schedule is given for pinworm?

MEDIUM

- A. Mebendazole 100 mg twice daily for 3 days
- B. Mebendazole 100 mg once, repeated after two weeks
- C. Praziquantel 5–10 mg/kg once
- D. Niclosamide 2 g over 2 days
- E. Pyrantel pamoate for 3 days

10. About what proportion of pinworm infections are asymptomatic?

MEDIUM

- A. Almost none
- B. One-third
- C. One-tenth
- D. Two-thirds
- E. All

11. A whole family is treated when one child has pinworm. Why?

MEDIUM

- A. Adults are always symptomatic
- B. The drug is ineffective in children alone
- C. Larvae migrate through the lungs
- D. The worm requires an intermediate host
- E. Eggs are widespread in household dust, bedding and surfaces and spread readily between members

12. An 8-year-old girl with pinworm has vaginal irritation and a mucoid discharge. What is the mechanism?

MEDIUM

- A. Worms crawling into the vulva and vagina
- B. Adult attachment to gastric mucosa
- C. Larval migration through the lungs
- D. Bacterial overgrowth from antibiotics
- E. Autoimmune vulvitis

13. Where do adult pinworms normally live?

MEDIUM

- A. Jejunum attached by buccal capsule
- B. Duodenal mucosa
- C. Sigmoid colon and rectum
- D. Bile duct
- E. Lumen of the cecum, appendix and adjacent ascending colon

14. Besides hand-to-mouth spread, which route is described for pinworm?

MEDIUM

- A. Skin penetration by filariform larvae
- B. Bite of an insect vector
- C. Undercooked beef
- D. Inhalation of airborne eggs in dust
- E. Trans-placental transmission

15. A child is treated but reinfected via bedsheets 10 days later. Which fact explains the persistent risk?

HARD

- A. Larvae survive 3–4 weeks in soil
- B. Eggs stay viable for about 2 weeks in cool moist conditions
- C. Adult females live for years
- D. Eggs need an intermediate host
- E. Eggs are killed only by freezing

16. A child sucks a thumb after scratching. Where do the swallowed eggs hatch?

HARD

- A. Small intestine
- B. Stomach
- C. Colon
- D. Lungs
- E. Perianal skin

17. A child with perianal itch has marked eosinophilia and high IgE. What is the most accurate interpretation?

HARD

- A. They confirm pinworm
- B. They indicate hookworm larvae only
- C. They confirm autoinfection
- D. These findings are not linked to enterobiasis, so look for another or coexisting cause
- E. They are typical of enterobiasis

18. An appendectomy specimen from a child shows adult pinworms. Which statement best fits the lecture?

HARD

- A. It proves hookworm infection
- B. It is a laboratory contaminant
- C. The worm inhabits the appendix and has been claimed to cause appendicitis
- D. It indicates tapeworm cysticercosis
- E. It shows a bile-duct infection

19. Why might a child with pinworm develop nocturnal enuresis?

HARD

- A. Larvae reach the bladder wall
- B. Night-time worm migration disturbs sleep
- C. Eosinophilic cystitis
- D. Kidney invasion
- E. Autoimmune bladder disease

20. Which combination best interrupts household transmission?

HARD

- A. Treating only the index child
- B. Freezing meat
- C. Wearing shoes outdoors
- D. Short nails, handwashing, cleaning surfaces and treating all household members
- E. Sanitary disposal of stool only

✓ Answer key

Q	Answer	Difficulty	Why
1	D	EASY	<i>E. vermicularis</i> causes enterobiasis, or pinworm infection.
2	E	EASY	Infection is common in children aged 5–14 years.
3	A	EASY	Eggs are collected from the pre-anal area with adhesive tape.
4	A	EASY	Migration of the gravid female causes pruritus ani.
5	C	MEDIUM	Oviposition at night is induced by cooler, aerobic conditions.
6	C	MEDIUM	The coiled embryo becomes infectious ~6 h after deposition (CDC figure says 4–6 h).
7	C	MEDIUM	Hookworm eggs are 60 × 40 µm with 7–8 blastomeres and a clear space; pinworm eggs are planoconvex, thick-shelled and float in salt.
8	E	MEDIUM	The double-bulb esophagus is unique to this worm, along with cervical alae.
9	B	MEDIUM	Hookworm uses 100 mg twice daily for 3 days; pinworm uses a single dose repeated at two weeks.
10	B	MEDIUM	About one-third of infections are asymptomatic.
11	E	MEDIUM	Household transmission is common; mass treatment reduces reinfection.
12	A	MEDIUM	Ectopic worm migration causes irritation and mucoid discharge.
13	E	MEDIUM	Adults inhabit the cecum, appendix and ascending colon; hookworms are in the jejunum.
14	D	MEDIUM	Airborne eggs in dust can be inhaled; retro-infection through the anus is rare.
15	B	HARD	Egg viability in the environment (~2 weeks) is why hygiene and repeat dosing matter.
16	A	HARD	On ingestion the eggs hatch in the small intestine; larvae molt twice and mature in about 36–53 days.
17	D	HARD	The lecture says eosinophilia and raised IgE are not associated with enterobiasis.
18	C	HARD	Adults live in the cecum/appendix region; a causal role in appendicitis has been claimed.
19	B	HARD	Nocturnal migration disturbs sleep, and enuresis is sometimes seen.
20	D	HARD	The lecture lists nail cutting, handwashing, surface cleaning, ointment and mass household treatment.